Idir Meradji

AI / ML Engineer

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Generative AI • Knowledge Graphs • Computer Vision • Probabilistic ML

Profile

Final-year MSc student in Data Science & AI at Université Paris-Saclay with industrial R&D experience in designing and deploying AI systems. My background spans machine learning, deep learning, generative AI, computer vision, knowledge representation, and data-driven applications, supported by strong software engineering skills. Seeking a full-time AI/ML Engineer role at the intersection of applied AI and production-grade engineering.

Skills

Programming	Python, Java, SQL, SPARQL, C, C++
Machine Learning & AI	Supervised & Unsupervised Learning, NLP, Generative AI, CNNs, Transfer Learning, Variational Autoencoders (VAE), Data Augmentation, PyTorch
LLM & MLOps Tooling	LangChain, LlamaIndex, vLLM, Ollama, MLflow, Weights & Biases
Knowledge Representation	Knowledge Graphs, RDF, OWL, SKOS, SHACL, SPARQL, Ontology Engineering
Data Science & Optimization	Feature Engineering, Statistical Learning, Bayesian Optimization, Gaussian Processes, Convex Optimization
Software & Data Engineering	REST APIs, Apache Spark, Hadoop, Git
Frameworks & Tools	PyTorch, TensorFlow, Scikit-learn, Hugging Face, FAISS, Protégé, Power BI, Streamlit, TorchVision

Professional Experience

AI / Software Engineer Intern
Dassault Systèmes – France *(Mar 2026 – Aug 2026)*
- Built an end-to-end GraphRAG assistant for natural-language exploration of ontologies, from the UI down to SPARQL execution over the knowledge graph.
- Built the ontology embedding pipeline in Java: RDF/SKOS extraction via SPARQL, batch vectorization through a Mistral embedding service, and an in-memory vector store with HNSW indexing, isolated per dataset.
- Developed the front-end panel (TypeScript, JavaScript, SCSS) inside the 3DEXPERIENCE Ontology Explorer: a conversational chat interface with a collapsible SPARQL-results table, wired to the GraphRAG API.
- Diagnosed and hardened multiple incompatibilities in the pipeline by resolving ~10 root-cause defects end-to-end.

AI Engineer Intern
BNP Paribas *(Mar 2025 – Jun 2025)*
- Developed a multi-agent RAG assistant for internal knowledge access, evaluating both LangChain and LlamaIndex for orchestration.
- Implemented semantic search over 20K+ internal documents using BGE multilingual embeddings, FAISS indexing, and a GPT model for answer generation.
- Built ML pipelines for document classification and information retrieval.

Data Analyst & NLP Intern
ThynkTech *(Nov 2024 – Jan 2025)*
- Built a document-clustering pipeline (TF-IDF + K-Means; Python, scikit-learn, spaCy) over 500+ documents to group content by theme.
- Surfaced the most-referenced resources and recurring topics, making the corpus significantly easier to navigate.

Machine Learning Intern
SAA – Société Nationale d'Assurance *(Mar 2024 – Jun 2024)*
- Developed a Variational Recurrent Autoencoder (VRAE) in TensorFlow to detect anomalies during data exploration over clients' historical records, reaching 94% accuracy.
- Delivered cleaner upstream data so downstream analysis and workflows ran on reliable inputs, cutting manual data-cleaning effort.

Education

Master of Science – Data Science (M2)
Paris-Saclay University, Faculty of Science *(Sep 2025 – Present)*
France Excellence IFA Scholarship recipient. Coursework: Advanced ML, Optimization, NLP, Probabilistic Graphical Models, Semantic Web, Data Mining.

Engineering & Master in Data Science and Artificial Intelligence
National Polytechnic School *(2022 – 2025)*
Graduated Top 5 in class, with High Honors. Thesis: "Hybrid Intelligent Assistant: A Multi-Agent Architecture for Contextual Reasoning and Information Retrieval." Relevant courses: Data Analysis, NLP, ML/DL, Big Data, BI, Distributed Systems, IoT.

Preparatory Classes – Science and Technology
National Polytechnic School *(2020 – 2022)*
Top 5% nationwide in entrance exam to Grandes Écoles. Relevant courses: Mathematics, Physics, Statistics, Programming, Economics.

Projects & Competitions

Hack4Future – Société Générale *(2024) (1st Prize – National)* - Designed an AI-assisted credit assessment workflow combining document analysis, consistency checking, and ML-based fraud detection to support financial risk decision-making.
Neural Feature Learning Study *(2026)* - Proposed a multi-model feature fusion approach combining deep feature extraction and representation learning to improve classification performance.
Brain Tumor Detection *(2026)* - Rebuilt as a CNN classifier that quantifies its own uncertainty (MC Dropout) and flags ambiguous scans for review; details in portfolio.
Academic Ecosystem Ontology *(2025)* - Designed an RDF/OWL ontology with semantic reasoning, SPARQL querying, and ontology engineering to enable inference and knowledge discovery across academic entities.
Transformer from Scratch *(2025)* - Implemented a Transformer architecture from scratch in PyTorch, including multi-head self-attention, positional encoding, and autoregressive text generation.
Entrepreneurship Project *(2026)* - Built and deployed a food planning application using recommendation algorithms, REST APIs, and user-centered design to promote healthy eating habits.
IEEE Hackathon *(2023) (2nd Place)* - Developed an intelligent student assistant integrating personalized recommendations, resource aggregation, and productivity services into a unified application.

Leadership & Activities

Human Resources Manager
IEC Club *(2024)*
- Led the recruitment and training of new members, coordinated large-scale events, and strengthened team engagement through leadership development initiatives.

Graphic Designer
IEC Club *(2023)*
- Created visual content for club communication and events, contributing to the club's branding and community visibility.

Languages

French (Native) • English (C1 – IELTS)